Internboot Project Report: Task Management Pro

1. Project Title

Task Management Pro with Real-time Firebase Integration.

2. Student Information

Name: Mannu Mehra

Degree: Bachelor of Computer Applications (BCA)

College: Arcade Business College (Patliputra University)

Internship: AI Data Quality Analyst / Web Development at Orbiqe Technologies (Internboot).

3. Introduction

This project is a modern, responsive To-Do application developed using the MERN (React) stack and Firebase. The goal was to create a secure environment where users can manage daily tasks with real-time updates.

4. Core Features

Google Authentication: Users sign in securely via Google, and the app displays their profile picture fetched from the Google API.

CRUD Functionality: Full support for Creating, Reading, Updating (Editing), and Deleting tasks.

Advanced Organization: Tasks are organized by Priority (High, Medium, Low) and Category (Work, Personal, Study).

Protected Routes: Navigation is restricted; users are redirected to the login page if they are not authenticated.

5. System Architecture

Frontend: Developed with React.js and Tailwind CSS for a responsive UI.

Backend/Database: Utilized Firebase Firestore for real-time NoSQL data storage.

State Management: Used React Context API to manage authentication states across the application.

6. Conclusion

Through this project, I gained hands-on experience in cloud integration, handling asynchronous operations in JavaScript, and managing complex application states.